Domenic J. Jernigan

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Education

University of California, San Diego La Jolla, CA

B.S. Cognitive Science, Spec. in Machine Learning & Neural Computation | Computer Science Minor | GPA: 3.6 *Exp. Jun 2027*

Technical Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL, JSON
ML & Data: Pandas, VS Code, Vercel, React, Next.js

Experience

Pedagogical Learning Apprentice Mar 2026 – Jun 2026

UCSD — COGS 108: Data Science in Practice *La Jolla, CA*

- Competitively selected by course faculty to support instruction in applied machine learning and data science across weekly sections and project reviews.
- Mentor students through end-to-end Python pipelines using pandas and scikit-learn — regression, classification, time-series forecasting, and unsupervised clustering.
- Employ Anthropic API to develop translational Claude Skill supporting the integration of Professor's visioned projects for future classes; converted class projects from Jupyter Notebooks to OtterGrader

Data Operations Specialist May 2025 – Present

UC San Diego, Office of Undergraduate Admissions *La Jolla, CA*

- Process high-volume applicant records — transcripts, test scores, correspondence — through scanning, indexing, and validation workflows in Slate CRM and OnBase.
- Maintain data integrity and FERPA compliance across Slate, OnBase, and Blade; reconcile discrepancies flagged during peak admissions cycles.
- Collaborate with Enrollment Management on workflow improvements supporting peak-cycle processing and continuous pipeline refinement.

Projects

EEG/sEMG Artifact Study | *Triton NeuroTech, Python, scikit-learn* Oct 2026 – Present

- Investigating whether low-cost EEG/sEMG hardware decodes genuine neural signal or movement artifacts (eye movement, facial muscle tension, head motion) during motor imagery tasks.
- Building artifact-only baselines and channel/frequency ablations to test how much decoding accuracy survives once neural signal is removed.
- Testing cross-session generalization of classifiers (e.g., CSP + LDA) to measure distribution shift between recordings.

Diabetes ML Research | *COGS 109, Python, scikit-learn, pandas* Oct 2026 – Dec 2026

- Built and compared 3 MLR classification models to predict Diabetes patient testing intensity from the UCI ML repo Diabetes 130-US Hospitals dataset (˜100K patient encounters).
- Handled class imbalance and high-cardinality clinical features (diagnosis codes, medications); evaluated with standard evaluation metrics to minimizes Mean Squared Error and inflates explained variance).
- Collaborated in a 3-person team to predict the number of lab procedures from different combinations of patient features (e.g., feature selection, hyper-parameter tuning, error analysis).

Crypto ML Project Lead | *COGS 108: Data Science in Practice* Jan 2026 – Mar 2026

- Led a peer team on an end-to-end ML study classifying Bitcoin as a financial instrument, extending peer-reviewed literature on cryptocurrency market behavior.
- Engineered features and trained regression, classification, and clustering models in scikit-learn; applied time-series forecasting to benchmark Bitcoin against equities, commodities, and currency indices.
- Owned scoping, task delegation, and experiment design across a 10-week project; authored the final report and presentation — earned an A in the course.